

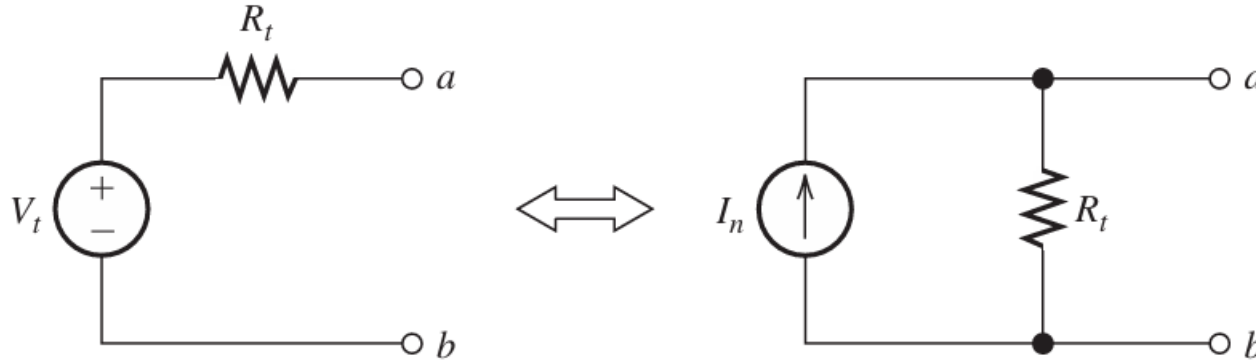
ES -116

Principles and Applications of Electrical Engineering

Semester II, 2025-26

Maximum Power Transfer and Energy Storage Elements

Summary of Thevenin and Norton Theorems

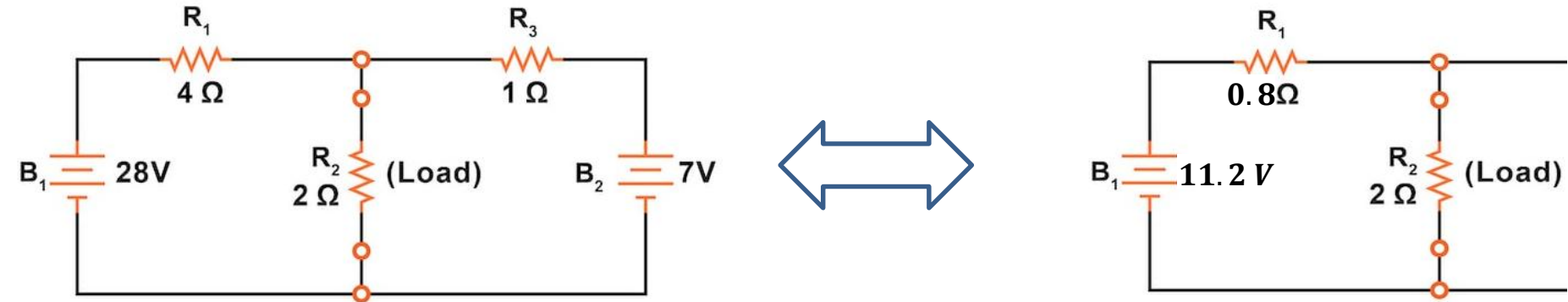


- Thévenin and Norton theorems let you replace any linear two-terminal network (resistors + voltage/current sources) by a simple equivalent seen from those terminals.
- Thévenin equivalent: one ideal voltage source V_t in series with a resistance R_t .
- Norton equivalent: one ideal current source I_n in parallel with a resistance R_t .
- The two are related by: $V_t = I_n R_t$
- Use them to simplify analysis when the load changes, so you don't re-solve the whole circuit each time.

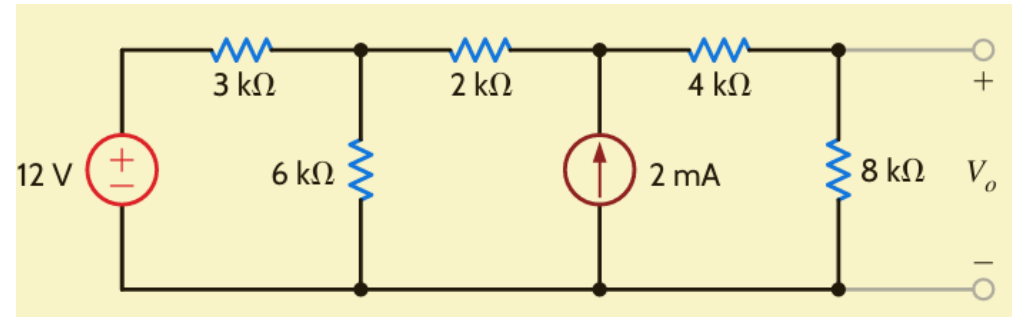
Recap: Thevenin/Norton Equivalent

- Simplifies the complex linear circuit to its Thevenin/Norton Equivalent.
- Helps us analyse the effect of the load on current drawn by (or voltage across) the load.

How do we check if the two networks are equivalent between node A and B?



Summary of Electrical Circuit Analysis Methods

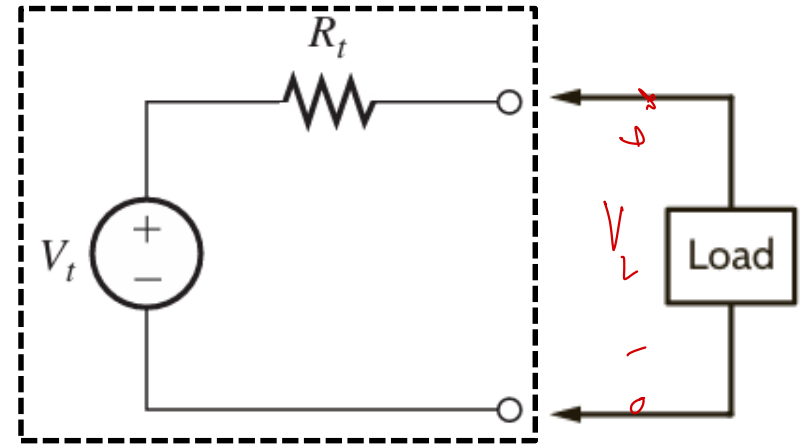
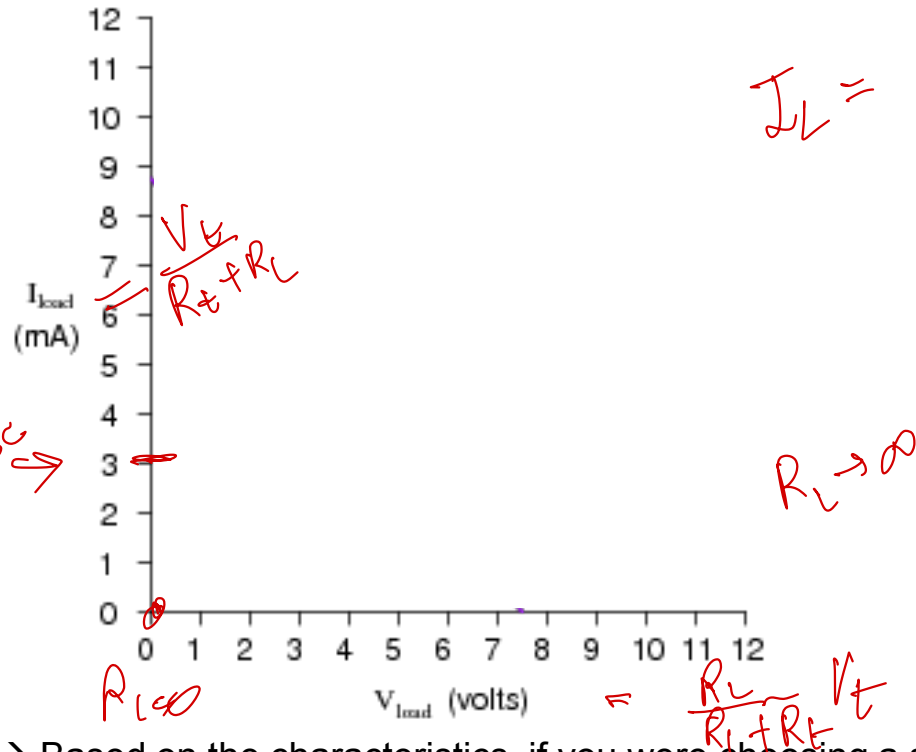


- What methods can you use to solve for V_o in the given circuit?

Practical Example for choice of load: Solar cell

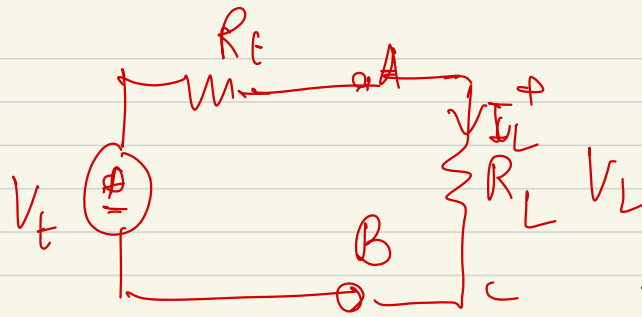
Q. How to get the maximum power transferred to the load?

Load Characteristics (V_{load} Vs I_{load})



Power Source:
 $V_t = 12 \text{ V}$, $R_t = 4 \Omega$

→ Based on the characteristics, if you were choosing a specific DC motor to run off this solar panel, would you choose a motor with a coil resistance of 1Ω , 4Ω , or 20Ω to get the most mechanical work out of it?



$$V_L \propto f(R_L)$$

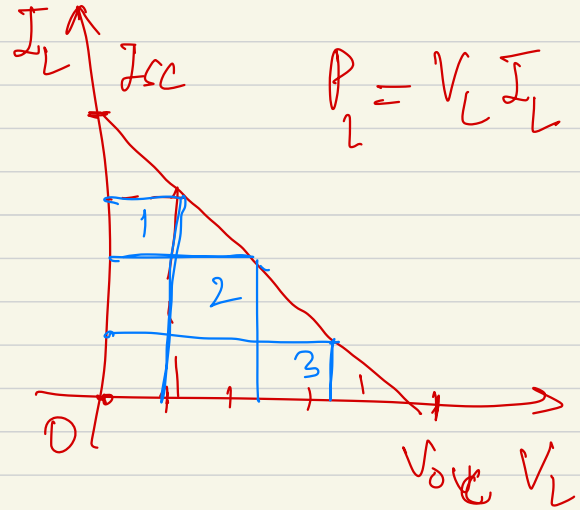
$$I_L \propto f(V_L)$$

$$P_L = V_L I_L$$

$$V_L = 0, \text{ when } R_L = 0, I_L = \frac{V_t}{R_t}$$

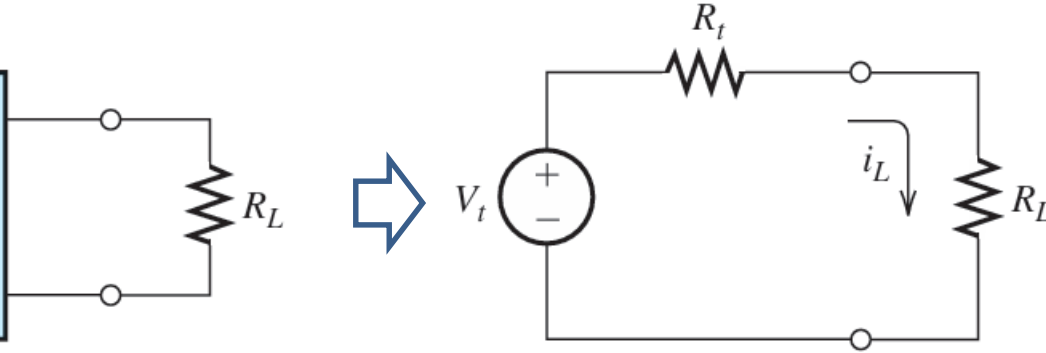
$$I_L = 0, \text{ when } R_L = \infty, V_L = V_t$$

$$0 < R_L < \infty$$



Maximum Power Transfer

Two-terminal
circuit of
sources and
resistances



Maximum Power Transfer:

$$\frac{dP_L}{dR_L} = 0$$

$$P_L = \frac{V_L^2}{R_L} = \left(\frac{R_L \cdot V_t}{R_L + R_t} \right)^2 \cdot \frac{1}{R_L}$$

$$R_L = R_t$$

Trivia: Can it help me understand the reason for the difficulty that I may go through in starting my car on a frigid morning?

Another Practical Scenario for Matching Resistance

The Scenario: Your Wi-Fi router ($R_{\text{Source}} = 50\Omega$) needs to send data to your antenna ($R_{\text{Load}} = 50\Omega$). You lost the correct cable, so you use an old TV cable ($R_{\text{Cable}} = 75\Omega$) to connect them.

1. Result: The antenna broadcasts a weaker radio wave. Your phone shows "1 Bar" instead of "Full Bars" because the signal physically doesn't reach as far.

Reason:

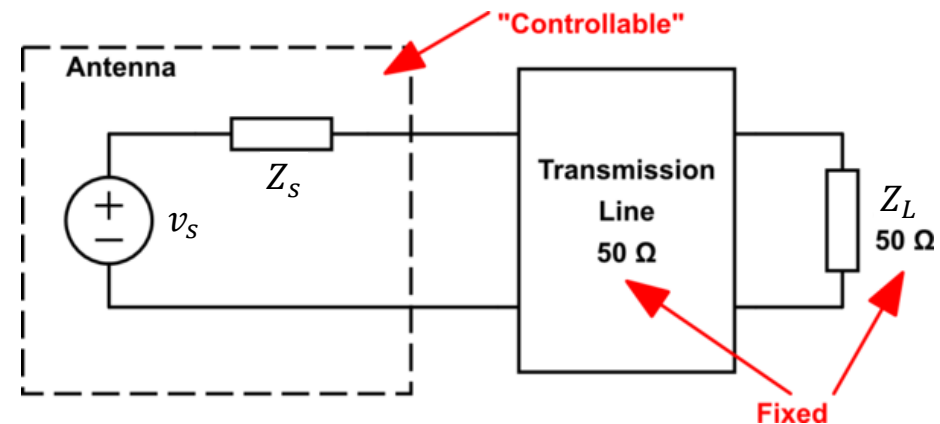
2. Result: You experience "electrical echoes" i.e. collision of reflected signals with new data coming out.

Effect: The router gets confused ("Did you say 'Hello' or 'He-He-llo-lo'?"). It has to pause and re-send the data.

Experience: This constant re-sending is what you experience as Lag or Buffering.

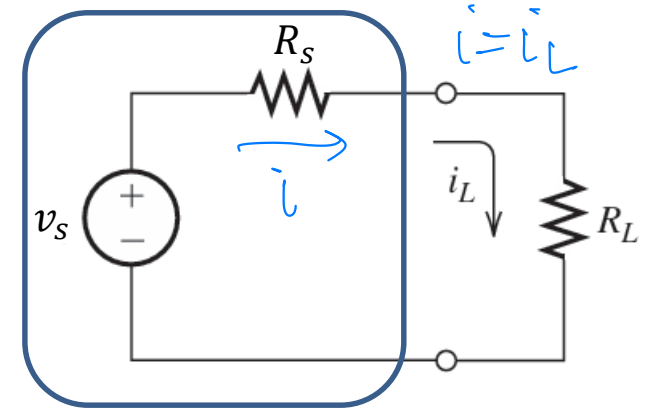
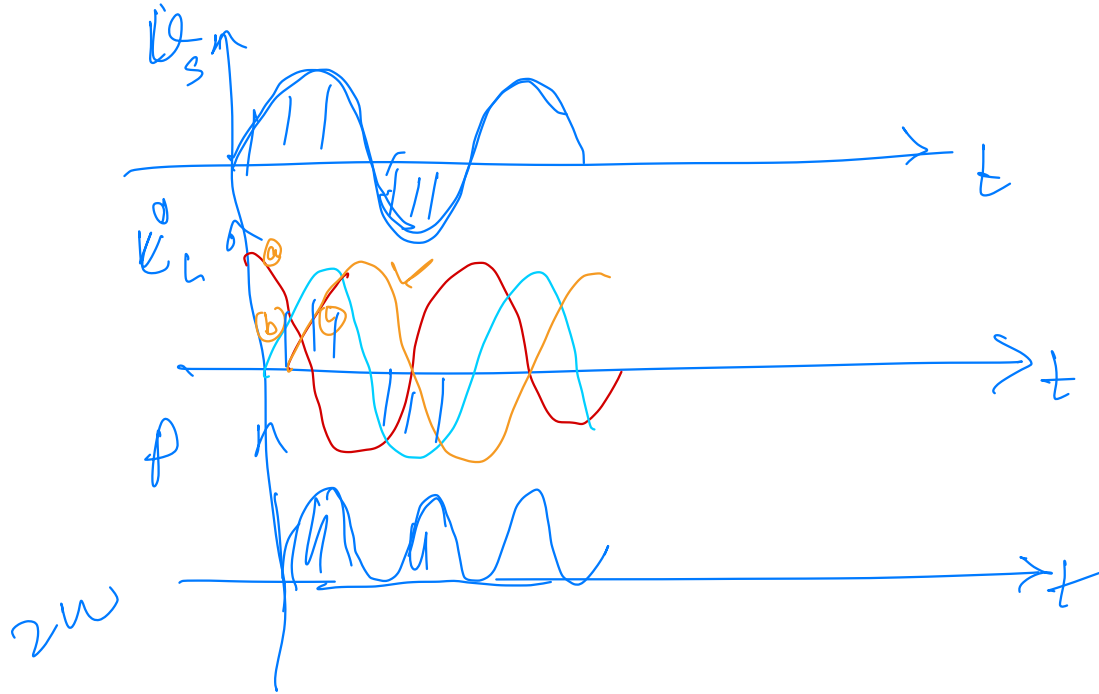
Reason:

Do you know the difference between resistance and impedance?



Reviewing Resistor Networks for AC input

How do current, voltage and power waveforms look like for the load?



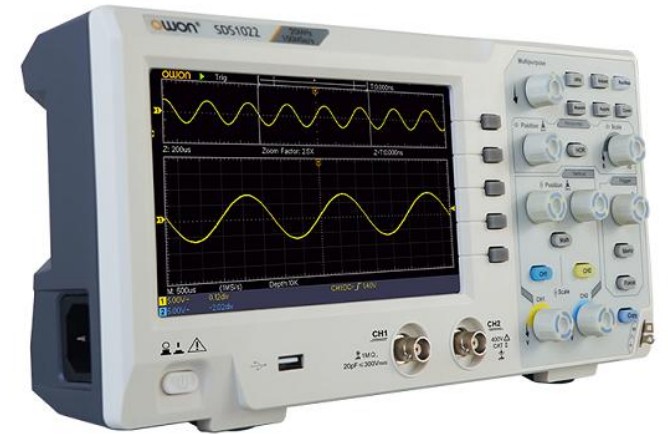
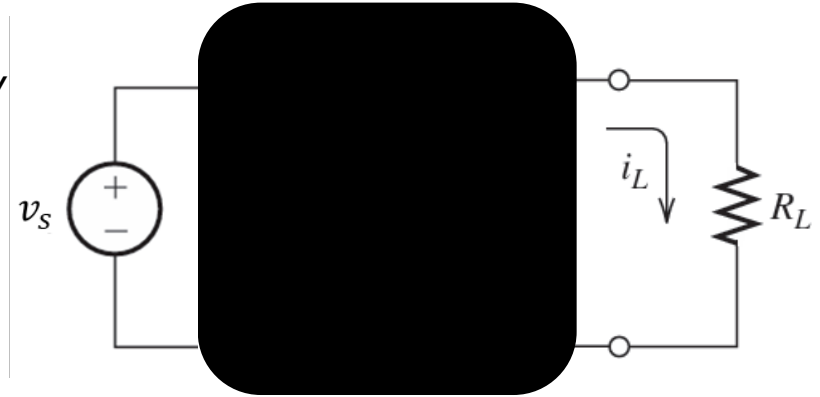
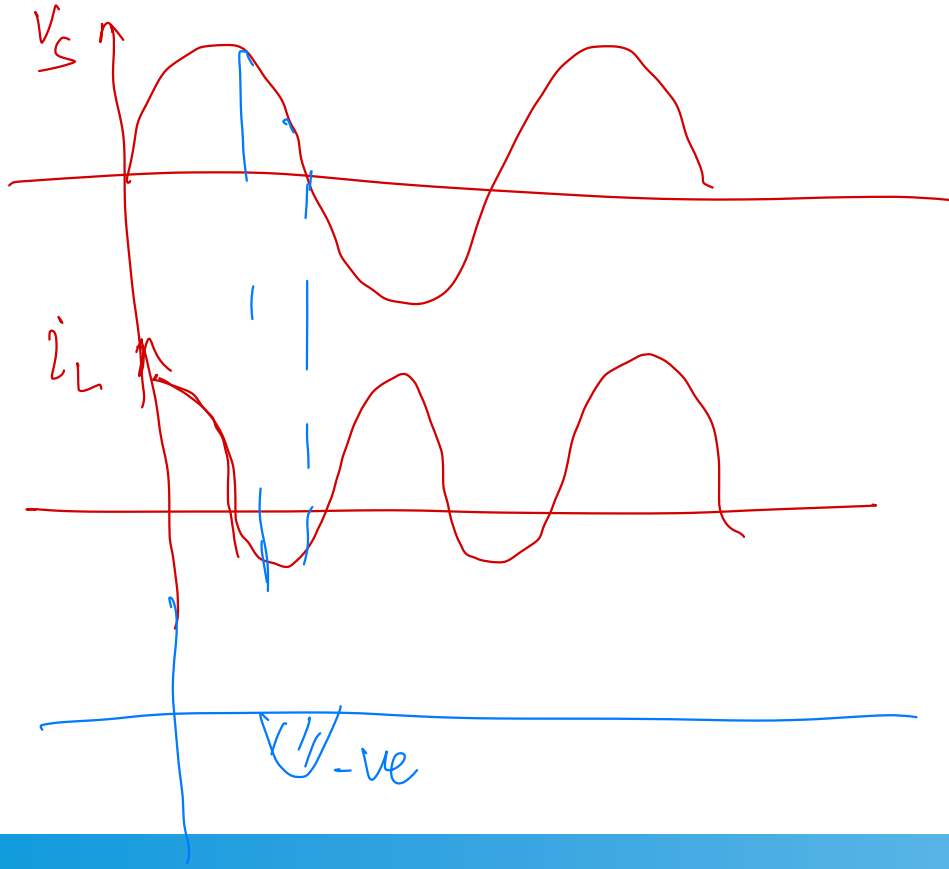
Blue box

$$V_s = i (R_s + R_L)$$

$$i(t) = \frac{V_s(t)}{R_s + R_L} = \frac{A \sin(\omega t)}{R_s + R_L}$$

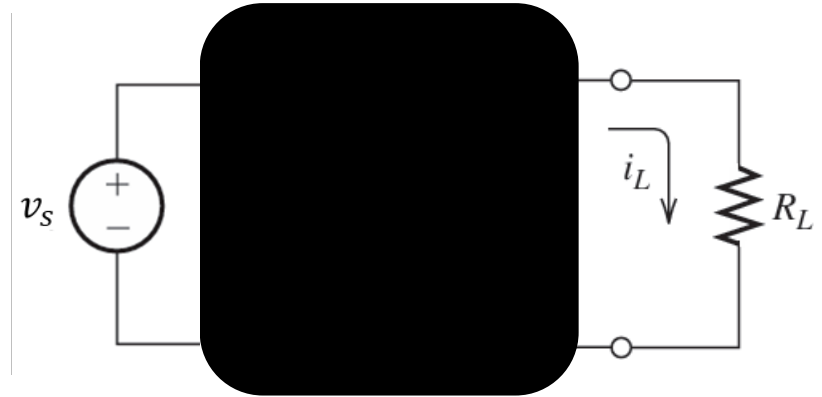
Anomaly observed for a black box

When you observe the Current through the load and voltage source on DSO, they are *not in phase (lagging or leading)*. Why



Forensics using Mathematics

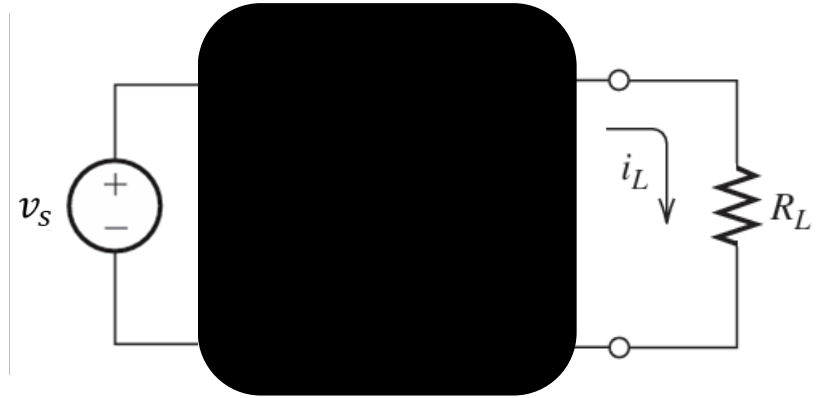
Can we calculate the power using the current and voltage waveforms?



Naming this new behaviour

We need a name for this 'borrowing and giving it back' behavior. It resists current, but not by friction (Resistance). It resists by reaction. We call it Reactance (X).

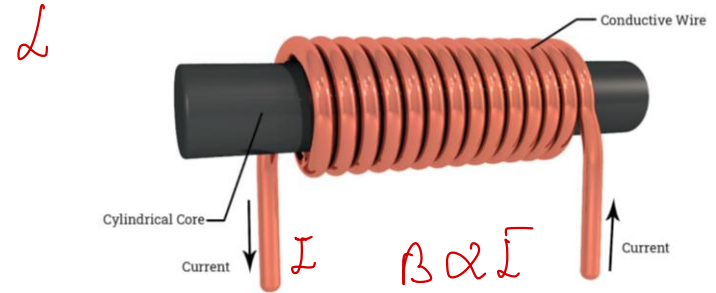
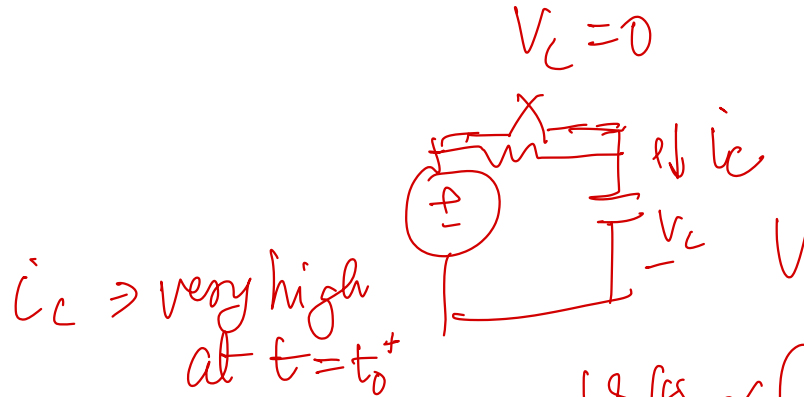
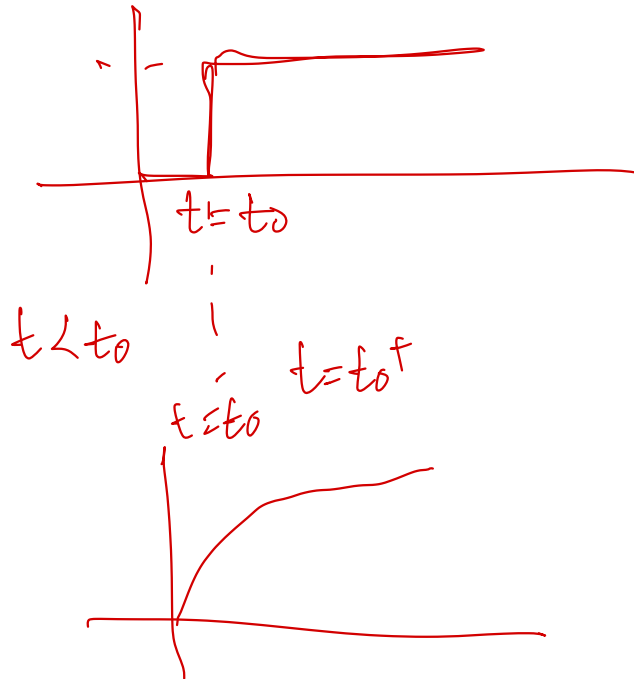
$$X_C = \frac{1}{\omega C}$$
$$X_L = \omega L$$



Electrical components that borrows energy and gives it back

- The Inductor (The Current Lagger)
- The Capacitor (The Current Leader)

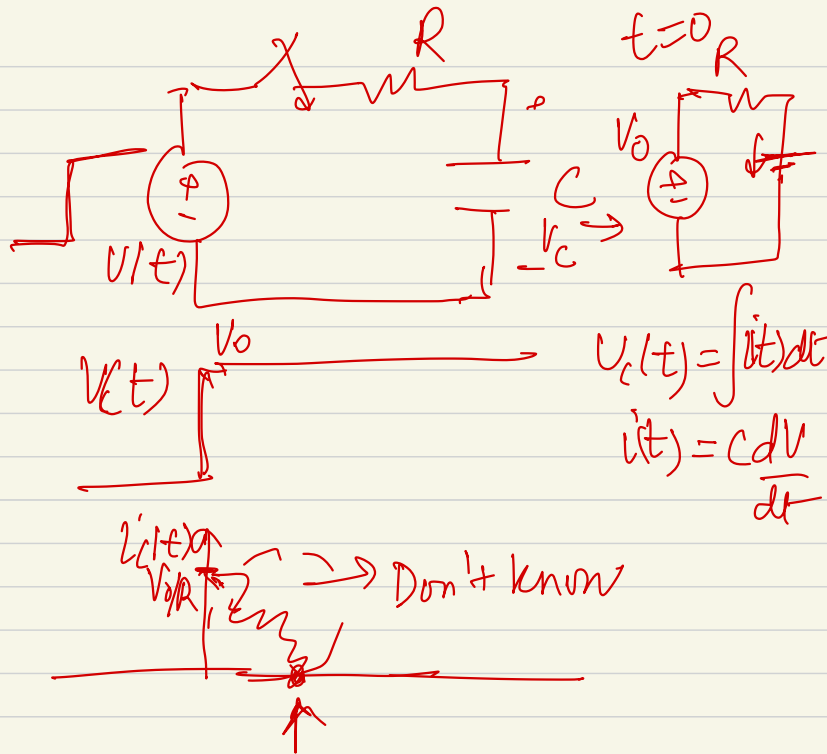
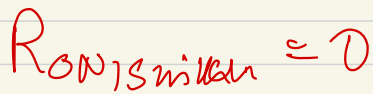
Can you think of a physical analogy for these components?



$$\Phi \propto I$$

$$\Phi \propto \frac{di}{dt}$$

$$V_c \propto \int i_c(t) dt \Rightarrow i_c = C \frac{dv_c}{dt}$$



In real systems, we experience a delay due to them!

Can you explain the delay in:

- 1) When you switch the fan off, motor doesn't stop instantly.
- 2) When you turn-on a LED/lamp, it takes a while to glow.

Electrical components that borrows energy and gives it back

- The Inductor (The Current Lagger)
- The Capacitor (The Current Leader)

Let us now understand their mathematically!

How does current relate to voltage for these components?

$$v = L \frac{di}{dt} \rightarrow i(t) = \frac{1}{L} \int v(t) dt$$
$$i = C \frac{dv}{dt} \rightarrow v(t) = \frac{1}{C} \int i(t) dt$$

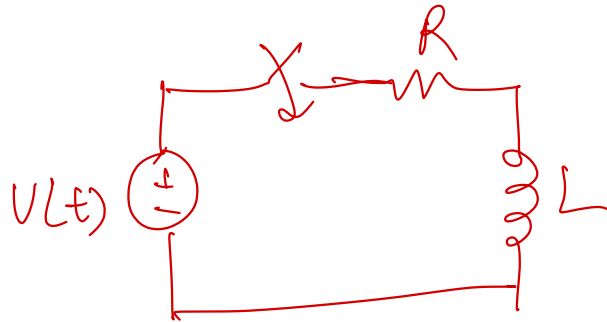
Response of these components in electrical circuits

We have two circuits connected to a 10V DC Switch. We close the switch at $t=0$.

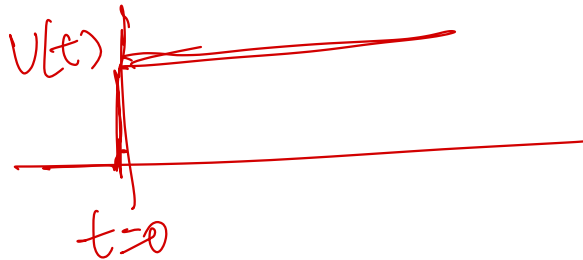
1. Circuit A (Reference): Just a Resistor ($1k\Omega$)
2. Circuit B (The Test): A Resistor ($1k\Omega$) in series with a Capacitor ($1000\mu F$) OR an Inductor ($100mH$).

Calculating the "Current inertia"

- The Inductor (The Current Lagger)
- Circuit B (The Test): A Resistor ($1\text{ k}\Omega$) in series with an Inductor (100 mH).



$$\text{At } t=0, i_L = 0, U_L = V(t)$$
$$V_L(t), i_L(t)$$



Calculating the delay

- The Capacitor (The Current Leader)
- Circuit B (The Test): A Resistor ($1\text{ k}\Omega$) in series with a Capacitor ($1000\text{ }\mu\text{F}$).